

Least-Coercive Competitive Assignment with Lower and Upper Congestion Bounds

A rigorous repair and partial answer to the lower-bound question of
Bogomolnaia and Moulin

Technical note

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Abstract

Bogomolnaia and Moulin ask how competitiveness should be defined when a lower congestion bound forces agents to populate a post that they unanimously dislike. A proposed “draft-and-compete” answer designates exactly the lower quota at each post as a noncompetitive mandatory block and applies ordinary competitiveness to the remaining agents. That proposal is not valid for the model with both lower and upper bounds: upper capacities at other posts can force more than the coordinatewise lower quota into a disliked post, and freezing mandatory identities can certify Pareto-dominated assignments.

This note gives a rigorous repair. We retain the original virtual-replacement quote $s_a \vee 1$ at every post. For a feasible assignment, an agent is competitively satisfied when her assigned post belongs to her demand at that quote. The *competitive deficit* is the number of agents who are not competitively satisfied, and a least-coercive assignment minimizes this deficit over all feasible assignments. For every fixed feasible congestion vector s , the minimum deficit is exactly a capacitated Hall deficiency. Consequently it has a transparent bottleneck certificate and is computable by one maximum-flow problem. We also prove that the minimum feasible population of any set of posts B is

$$\beta(B) = \max\{\ell(B), n - u(\mathcal{A} \setminus B)\},$$

which shows precisely how upper bounds outside B create additional compulsory service inside B . In the two-post unanimously loathed case, the least-coercive deficit equals this bound. A fractional analogue always exists, has the same Hall formula at a fixed expected congestion vector, and admits a floor/ceiling lottery implementation.

The construction is an exact extension of ordinary competitiveness whenever zero deficit is feasible, but it does not preserve the original uniqueness or efficiency conclusions and it is not uniquely characterized by axioms. It is therefore *partial progress, not a complete solution* of the authors’ normative open problem.

1 Status and motivation

Bogomolnaia and Moulin define a competitive assignment by treating congestion as a post-specific price: an agent assigned to post a at congestion s_a must weakly prefer (a, s_a) to $(x, s_x \vee 1)$ at every post x ; an empty post is quoted at congestion one [1]. They show that, when such a deterministic assignment exists, it has strong uniqueness, welfare, and fairness properties. In their concluding section they ask how the definition should change under lower and upper bounds, especially when a positive lower bound forces agents into a unanimously loathed post.

The difficulty is not merely that a particular proof stops working. A lower bound can make the all-agent competitive requirement logically infeasible. Any repair must therefore say which violations are genuinely forced and must do so using the entire system of lower and upper bounds. A fixed exemption of exactly ℓ_a agents at post a does not accomplish this: upper capacities elsewhere can force additional agents into a .

The approach below keeps the competitive comparison itself unchanged and makes the smallest possible relaxation in a literal optimization sense: minimize the number of agents for whom the competitive condition fails. Minimizing the number of envy or stability violations is also a standard response to lower-quota infeasibility in other matching models; see, for example, Hu and Garg [2]. The present model is one-sided and congestion-dependent, and the resulting fixed-load problem has a particularly clean Hall-deficiency characterization.

Status. The results below are a rigorous partial answer. They correct the mathematical errors in the fixed-draft proposal and produce a well-defined, always-existing relaxation with exact structural and fractional results. They do not establish that this relaxation is the uniquely correct normative definition of competitiveness.

2 Model and the original competitive comparison

Let $\mathcal{A}$ be a finite set of posts and $\mathcal{N}$ a finite set of agents, with $|\mathcal{N}| = n \geq 1$. Each post a has integer lower and upper bounds

$$0 \leq \ell_a \leq u_a \leq n, \quad \sum_{a \in \mathcal{A}} \ell_a \leq n \leq \sum_{a \in \mathcal{A}} u_a.$$

These conditions imply that at least one feasible integer congestion vector exists. Write

$$\mathcal{A}^+ := \{a \in \mathcal{A} : u_a \geq 1\}.$$

A deterministic assignment is a partition $P = (S_a)_{a \in \mathcal{A}}$ of $\mathcal{N}$, with congestion vector $s(P) = (s_a)_{a \in \mathcal{A}}$ defined by $s_a = |S_a|$. It is feasible if

$$\ell_a \leq s_a \leq u_a \quad (a \in \mathcal{A}).$$

The set of feasible congestion vectors is

$$\mathcal{S} := \left\{ s \in \mathbb{Z}_+^{\mathcal{A}} : \ell \leq s \leq u, \sum_{a \in \mathcal{A}} s_a = n \right\}.$$

Agent i has a complete and transitive preference relation $\succeq_i$ over allocations (a, k) , strictly decreasing in congestion at a fixed post:

$$(a, k) \succ_i (a, k+1) \quad (a \in \mathcal{A}^+, 1 \leq k < n).$$

For a feasible congestion vector s , retain the original virtual-replacement quote

$$q_a(s) := s_a \vee 1 = \max\{s_a, 1\}, \quad a \in \mathcal{A}^+.$$

The competitive demand of agent i is

$$D_i(s) := \{a \in \mathcal{A}^+ : (a, q_a(s)) \succeq_i (b, q_b(s)) \text{ for every } b \in \mathcal{A}^+\}. \quad (1)$$

The set is nonempty. A feasible assignment P is *bounded-competitive* if

$$i \in S_a \implies a \in D_i(s(P)). \quad (2)$$

When all bounds are vacuous, this is exactly Definition 3 of Bogomolnaia and Moulin. With upper bounds, the comparison to a full post remains a virtual replacement of an incumbent and therefore leaves its congestion unchanged. A lower bound does not alter this physical quote; it may instead make (2) impossible.

3 Why a fixed lower-quota draft is not a valid repair

This section isolates the two central defects of the fixed-draft construction.

Proposition 3.1 (The original demand condition cannot always be met). *For every integer $q \geq 1$, there is a feasible bounded instance in which every assignment violates (2) for exactly q agents.*

Proof. Take two posts a, b , $n = q + 1$ agents, and exact bounds

$$\ell_a = u_a = q, \quad \ell_b = u_b = 1.$$

Let every agent strictly prefer $(b, 1)$ to (a, q) . The unique feasible congestion vector is $(q, 1)$, and every agent's competitive demand is $\{b\}$. The one agent assigned to b satisfies (2); the q agents assigned to a do not. No all-agent version of the original demand condition can therefore be universally feasible. $\square$

The preceding proposition justifies a relaxation, but not an exemption of exactly the coordinate-wise lower quota. Upper capacities at other posts can create additional forced occupancy.

Proposition 3.2 (Exactly ℓ_a exemptions need not suffice). *There is a two-post instance for which no assignment can satisfy the fixed-draft condition that exactly ℓ_a occupants of each post are exempt and every other occupant must demand her post at the shifted quote*

$$\ell_x + \max\{s_x - \ell_x, 1\}.$$

Proof. Let $n = 3$ and

$$(\ell_a, u_a) = (1, 3), \quad (\ell_b, u_b) = (0, 1).$$

For every agent, choose monotone preferences with

$$(b, 1) \succ_i (a, 2) \succ_i (a, 3).$$

The only feasible congestion vectors are $(2, 1)$ and $(3, 0)$. At $(2, 1)$ the shifted quotes are $(a, 2)$ and $(b, 1)$, so neither occupant of a demands a . Exempting exactly $\ell_a = 1$ of them leaves one violation. At $(3, 0)$ the shifted quotes are $(a, 3)$ and $(b, 1)$, so all three occupants of a reject a and two violations remain after the single exemption. Hence no fixed-draft outcome exists. $\square$

There is a second problem with the shifted quote. It freezes the identity of the mandatory occupants and evaluates a possible replacement at congestion $\ell_a + 1$ rather than at the actual congestion ℓ_a .

Example 3.3 (A fixed draft can certify a Pareto-dominated assignment). Let $n = 2$, $\mathcal{A} = \{a, b\}$, and

$$(\ell_a, u_a) = (1, 2), \quad (\ell_b, u_b) = (0, 2).$$

Consider the assignment P in which agent 1 is assigned to a and designated mandatory, while agent 2 is assigned to b . Complete the following rankings monotonically:

$$\begin{aligned} 1 : & (b, 1) \succ_1 (a, 1) \succ_1 (b, 2) \succ_1 (a, 2), \\ 2 : & (a, 1) \succ_2 (b, 1) \succ_2 (a, 2) \succ_2 (b, 2). \end{aligned}$$

The shifted residual quotes at P are $(a, 2)$ and $(b, 1)$, so the only discretionary agent, agent 2, demands b . Thus P passes the fixed-draft test. But swapping the two agents gives agent 1 the allocation $(b, 1)$ and agent 2 the allocation $(a, 1)$, strictly improving both. At the actual virtual-replacement quote $(a, 1)$, agent 2 correctly expresses willingness to take over the lower-quota duty.

The valid part of the fixed-draft idea is only conditional: once a mandatory block is fixed exogenously and declared nontransferable, the residual problem is an ordinary congested-assignment problem with shifted preferences. That reduction does not show that the block has the correct size or identities, nor does it transfer efficiency or welfare conclusions to the mandatory agents.

4 Least-coercive competitiveness

We now make the set of competitive violations endogenous.

Definition 4.1 (Competitive deficit). For a feasible assignment $P = (S_a)_{a \in \mathcal{A}}$ with congestion vector $s = s(P)$, define its set of competitively satisfied agents by

$$C(P) := \{i \in \mathcal{N} : i \in S_a \text{ for some } a \in D_i(s)\}.$$

The *competitive deficit* of P is

$$\delta(P) := n - |C(P)|.$$

Equivalently, $\delta(P)$ is the number of agents whose assigned post is outside their competitive demand.

Definition 4.2 (Least-coercive competitive assignment). Set

$$\delta^* := \min\{\delta(P) : P \text{ is feasible}\}.$$

A feasible assignment is *least-coercive competitive* if $\delta(P) = \delta^*$.

Existence is immediate because there are finitely many feasible assignments. The substantive content is that the deficit has an exact bottleneck characterization. For a fixed $s \in \mathcal{S}$, write

$$\delta(s) := \min\{\delta(P) : s(P) = s\}.$$

For $X \subseteq \mathcal{N}$, let

$$\Gamma_s(X) := \bigcup_{i \in X} D_i(s), \quad s(B) := \sum_{a \in B} s_a.$$

For $B \subseteq \mathcal{A}^+$, also define

$$I_s(B) := \{i \in \mathcal{N} : D_i(s) \subseteq B\}.$$

Theorem 4.3 (Capacitated Hall deficiency). *For every feasible congestion vector s ,*

$$\delta(s) = \max_{X \subseteq \mathcal{N}} \{ |X| - s(\Gamma_s(X)) \} = \max_{B \subseteq \mathcal{A}^+} \{ |I_s(B)| - s(B) \}. \quad (3)$$

The empty set is allowed, so both maxima are nonnegative. A minimum-deficit assignment with congestion s is obtained by one capacitated maximum-flow problem.

Proof. Form the bipartite demand graph with agent vertices $i \in \mathcal{N}$, post vertices $a \in \mathcal{A}^+$, and edge (i, a) exactly when $a \in D_i(s)$. Give every agent capacity one and post a capacity s_a . Let $\nu(s)$ be the maximum cardinality of a capacitated matching in this graph.

For any assignment with congestion s , its competitively satisfied agents and their assigned edges form such a matching. Hence it has at most $\nu(s)$ satisfied agents and at least $n - \nu(s)$ violations. Conversely, take a maximum demand matching. It assigns $\nu(s)$ agents to demanded posts without exceeding any s_a . There are $n - \nu(s)$ unmatched agents and exactly the same number of unfilled seats in total. Because every agent can physically be assigned to every post, assign the unmatched agents arbitrarily to those seats. This completes the matching to an assignment with congestion s and exactly $n - \nu(s)$ violations. Therefore

$$\delta(s) = n - \nu(s).$$

Represent the capacitated matching as a flow network: the source has a unit-capacity arc to every agent, every demand edge has infinite capacity, and post a has an arc of capacity s_a to the sink. A finite source-sink cut is determined by a set $X \subseteq \mathcal{N}$ of agents retained on the source side and must also retain every post in $\Gamma_s(X)$. Its capacity is

$$n - |X| + s(\Gamma_s(X)).$$

The max-flow/min-cut theorem [3] gives

$$\nu(s) = \min_{X \subseteq \mathcal{N}} \{ n - |X| + s(\Gamma_s(X)) \},$$

which is the first equality in (3).

For the second equality, given X set $B = \Gamma_s(X)$. Then $X \subseteq I_s(B)$, so

$$|X| - s(\Gamma_s(X)) \leq |I_s(B)| - s(B).$$

Conversely, given B , set $X = I_s(B)$. Since $\Gamma_s(X) \subseteq B$,

$$|I_s(B)| - s(B) \leq |X| - s(\Gamma_s(X)).$$

Taking maxima proves the claim. □

The first formula says that a group X cannot all be competitively satisfied if the posts they collectively demand contain too few seats. The second says that if too many agents insist on choosing within a set B , some must be assigned outside B . Thus every unit of competitive deficit has a Hall-type scarcity certificate.

Corollary 4.4 (Exact consistency). *For a feasible congestion vector s , the following are equivalent:*

- (i) $\delta(s) = 0$;

(ii) $|X| \leq s(\Gamma_s(X))$ for every $X \subseteq \mathcal{N}$;

(iii) there is a bounded-competitive assignment with congestion s .

Consequently, $\delta^* = 0$ if and only if a bounded-competitive assignment exists, and in that case the least-coercive assignments are exactly the bounded-competitive assignments.

Proof. The equivalence of (i) and (ii) is Theorem 4.3; the equivalence of (i) and (iii) follows from the definition of $\delta(s)$. Minimizing over $s \in \mathcal{S}$ gives the final statement. $\square$

Corollary 4.5 (An exact algorithm for fixed number of posts). *The least-coercive deficit and an attaining assignment can be computed by enumerating $s \in \mathcal{S}$ and solving one maximum-flow problem for each s . In particular, for a fixed number of posts this is polynomial in n and in the size of the explicit preference table.*

Proof. There are at most $(n+1)^{|\mathcal{A}|-1}$ nonnegative integer vectors summing to n . For each feasible vector, compute the demand graph and apply Theorem 4.3. Retain a vector with minimum deficit and extend a maximum demand matching as in the theorem's proof. $\square$

5 What the full system of bounds actually forces

The fixed-draft proposal treats ℓ_a as the number of unavoidable occupants at post a . The correct feasibility calculation uses upper bounds at all other posts as well, and it naturally applies to sets of posts. For $B \subseteq \mathcal{A}$, write

$$\ell(B) := \sum_{a \in B} \ell_a, \quad u(B) := \sum_{a \in B} u_a.$$

Theorem 5.1 (Exact forced population of a set of posts). *For every $B \subseteq \mathcal{A}$,*

$$\boxed{\min_{s \in \mathcal{S}} s(B) = \beta(B) := \max\{\ell(B), n - u(\mathcal{A} \setminus B)\}.} \quad (4)$$

Proof. Every feasible s satisfies $s(B) \geq \ell(B)$. It also satisfies

$$s(B) = n - s(\mathcal{A} \setminus B) \geq n - u(\mathcal{A} \setminus B),$$

so $s(B) \geq \beta(B)$.

Set $t = \beta(B)$. Global feasibility implies

$$\ell(B) \leq t \leq u(B) \quad \text{and} \quad \ell(\mathcal{A} \setminus B) \leq n - t \leq u(\mathcal{A} \setminus B).$$

For integer lower and upper bounds, every integer total between the sum of the lower bounds and the sum of the upper bounds can be attained: start at the lower-bound vector and distribute the required remaining units without exceeding the upper bounds. Apply this observation separately on B with total t and on $\mathcal{A} \setminus B$ with total $n - t$. Combining the two vectors gives a feasible s with $s(B) = t$. $\square$

For a singleton, this becomes

$$\min_{s \in \mathcal{S}} s_a = \max \left\{ \ell_a, n - \sum_{x \neq a} u_x \right\}. \quad (5)$$

The second term is exactly what the fixed-draft proposal misses.

Corollary 5.2 (A unanimously undemanded set). *Fix $B \subseteq \mathcal{A}$. If at a feasible congestion vector s no agent demands any post in B , then every assignment with congestion s has at least $s(B)$ competitive violations. If this no-demand property holds for every $s \in \mathcal{S}$, then*

$$\delta^* \geq \beta(B).$$

Proof. All occupants of B are assigned outside their demand, so $\delta(P) \geq s(B)$. The second statement follows from Theorem 5.1 after minimizing over feasible s . Equivalently, use Theorem 4.3 with the set of posts $\mathcal{A} \setminus B$ that contains every agent's demand. $\square$

The bound is exact in the motivating two-post case.

Corollary 5.3 (Exact solution of the two-post loathing calculation). *Let $\mathcal{A} = \{a, b\}$ and suppose every agent has*

$$D_i(s) = \{b\} \quad \text{for every } s \in \mathcal{S}.$$

Then

$$\delta^* = \max\{\ell_a, n - u_b\}. \quad (6)$$

Proof. The lower bound follows from Corollary 5.2. By Theorem 5.1, choose a feasible vector with

$$s_a = \max\{\ell_a, n - u_b\}, \quad s_b = n - s_a.$$

At this vector all agents demand only b , so a maximum demand matching assigns exactly s_b agents competitively to b . The remaining s_a agents are assigned to a , giving deficit s_a and attaining the lower bound. $\square$

In Proposition 3.2, formula (6) gives

$$\delta^* = \max\{1, 3 - 1\} = 2.$$

Thus the true unavoidable burden is two agents, not the lower quota one.

6 Fractional least-coercive competition

Bogomolnaia and Moulin obtain unconditional existence after passing to a fractional expected congestion vector in the unbounded model. With hard bounds, zero-deficit fractional competition may again be infeasible. The least-coercive relaxation still has unconditional existence and an exact fixed-profile formula.

Give each agent a cardinal utility $v_i(a, k)$ representing $\succeq_i$ at integer congestion levels, and extend it linearly between consecutive integers. Linear interpolation is important if $v_i(a, t)$ is to equal expected utility when realized congestion rounds t up or down. For a feasible expected congestion vector

$$\sigma \in \Sigma := \left\{ z \in \mathbb{R}_+^{\mathcal{A}} : \ell \leq z \leq u, \sum_a z_a = n \right\},$$

define

$$D_i(\sigma) := \arg \max_{a \in \mathcal{A}^+} v_i(a, \sigma_a \vee 1).$$

A feasible fractional assignment is a matrix $x = (x_{ia})$ satisfying

$$x_{ia} \geq 0, \quad \sum_a x_{ia} = 1, \quad \sigma_a(x) := \sum_i x_{ia} \in [\ell_a, u_a].$$

Its competitive mass and fractional deficit are

$$V(x) := \sum_{i \in \mathcal{N}} \sum_{a \in D_i(\sigma(x))} x_{ia},$$

$$\Delta(x) := n - V(x).$$

A fractional assignment is *fractionally least-coercive* if it minimizes $\Delta(x)$.

Theorem 6.1 (Fractional existence, Hall formula, and implementation). *The following statements hold.*

- (i) *A fractionally least-coercive assignment exists.*
- (ii) *For a fixed $\sigma \in \Sigma$, the minimum fractional deficit among matrices with column sums σ is*

$$\Delta(\sigma) = \max_{X \subseteq \mathcal{N}} \{ |X| - \sigma(\Gamma_\sigma(X)) \}, \quad (7)$$

where $\Gamma_\sigma(X) = \bigcup_{i \in X} D_i(\sigma)$ and $\sigma(B) = \sum_{a \in B} \sigma_a$.

- (iii) *If Δ^* denotes the globally minimum fractional deficit, then $\Delta^* \leq \delta^*$. For every integral feasible vector s , $\Delta(s) = \delta(s)$.*
- (iv) *Let x be any fractionally least-coercive assignment and put $\sigma = \sigma(x)$. Then x is a convex combination of feasible deterministic assignments whose congestion at every post a is either $\lfloor \sigma_a \rfloor$ or $\lceil \sigma_a \rceil$, and every agent-post edge used in the lottery belongs to the support of x . In particular, if $\Delta^* = 0$, every realized assignment sends each agent only to a post in $D_i(\sigma)$.*

Proof. Existence. The feasible set of fractional matrices is nonempty and compact. We show that V is upper semicontinuous. Let $x^k \rightarrow x$ and pass to a subsequence along which $V(x^k)$ converges to its limsup and, for every agent i , the finite set $D_i(\sigma(x^k))$ is constant, say equal to E_i . Continuity of the utilities implies $E_i \subseteq D_i(\sigma(x))$: every weak maximizing inequality along the subsequence survives in the limit. Hence

$$\limsup_{k \rightarrow \infty} V(x^k) = \sum_i \sum_{a \in E_i} x_{ia} \leq \sum_i \sum_{a \in D_i(\sigma(x))} x_{ia} = V(x).$$

Thus V attains a maximum, equivalently Δ attains a minimum.

Fixed-profile formula. Fix σ . In the bipartite demand graph, give each agent capacity one and post a capacity σ_a . The maximum amount of flow sent through demand edges is some $\nu(\sigma)$. Any such demand flow can be extended to a full fractional assignment with column sums σ : route the residual agent supplies to the residual post capacities through unrestricted agent-post edges. Therefore $\Delta(\sigma) = n - \nu(\sigma)$. The same min-cut argument as in Theorem 4.3, now with real post capacities, gives (7).

Comparison with deterministic assignments. Every deterministic assignment is a feasible fractional matrix, so $\Delta^* \leq \delta^*$. If $\sigma = s$ is integral, the demand-flow network has integral capacities and therefore an integral maximum flow. The fixed-profile fractional and deterministic deficits coincide.

Implementation. Consider the polytope of matrices z satisfying

$$\sum_a z_{ia} = 1, \quad \lfloor \sigma_a \rfloor \leq \sum_i z_{ia} \leq \lceil \sigma_a \rceil, \quad z_{ia} = 0 \text{ if } x_{ia} = 0, \quad z_{ia} \geq 0.$$

It contains x . Its constraint matrix is a bipartite network matrix with integral right-hand sides, hence is totally unimodular [3]. Every extreme point is integral, so x is a convex combination of integral extreme points. At an integral extreme point each row contains one unit, each column sum is a floor or ceiling of σ_a , and only support edges of x are used. Because ℓ_a, u_a are integers and $\sigma_a \in [\ell_a, u_a]$, these rounded column sums respect the original bounds. If $\Delta^* = 0$, all positive entries of x are demand edges, so the same holds in every deterministic component. $\square$

The implementation statement evaluates demand at the common expected quote σ , exactly as in the fractional model of Bogomolnaia and Moulin. It does not claim that every realized assignment is least-coercive relative to its own realized congestion vector.

7 Sharing the unavoidable burden ex ante

The least-coercive criterion determines the smallest number of violations but may not determine which agents bear them. A lottery can share this burden without declaring an arbitrary lower-quota block in advance.

Let $\mathcal{L}^*$ be the finite set of least-coercive deterministic assignments. For $P \in \mathcal{L}^*$, define the violation vector $f(P) \in \{0, 1\}^{\mathcal{N}}$ by

$$f_i(P) = \mathbf{1}\{i \notin C(P)\}.$$

Every such vector has coordinate sum δ^* . The set of attainable ex ante violation-probability vectors is

$$\mathcal{B}^* := \text{conv}\{f(P) : P \in \mathcal{L}^*\}.$$

Proposition 7.1 (A balanced burden lottery). *There is a unique vector*

$$p^* \in \arg \min_{p \in \mathcal{B}^*} \sum_{i \in \mathcal{N}} p_i^2. \quad (8)$$

It is implementable by a lottery over least-coercive assignments. Moreover, every permutation of agents that preserves the preference profile also preserves p^ . If all agents have identical preferences, then*

$$p_i^* = \frac{\delta^*}{n} \quad (i \in \mathcal{N}).$$

Proof. The set $\mathcal{B}^*$ is nonempty, compact, and convex. The objective in (8) is strictly convex, so its minimizer is unique. By the definition of a convex hull, p^* is generated by a finite lottery over $\mathcal{L}^*$.

A preference-preserving permutation maps least-coercive assignments to least-coercive assignments and therefore maps $\mathcal{B}^*$ to itself. The objective is permutation invariant. Uniqueness then forces the minimizer to be invariant under every such permutation. If all preferences coincide, all agent permutations are symmetries, so every coordinate of p^* is equal. Their sum is δ^* , giving the stated formula. $\square$

This quadratic balancing rule is only one possible secondary fairness criterion. Its role is to show that equal liability can be imposed on the *endogenously unavoidable* burden, rather than on an exogenously chosen block of size $\sum_a \ell_a$.

8 Limitations and why the result is only partial progress

Least-coercive competitiveness restores exact consistency and gives a sharp measure of unavoidable competitive failure, but it does not inherit the strong conclusions of the unbounded zero-deficit model.

Example 8.1 (No congestion uniqueness, welfare uniqueness, or efficiency). Let $n = 2$, let $\mathcal{A} = \{a, b, c\}$, and impose unit upper capacities and zero lower bounds:

$$\ell_x = 0, \quad u_x = 1 \quad (x \in \{a, b, c\}).$$

Both agents have identical strict rankings of the relevant unit-congestion allocations,

$$(a, 1) \succ (b, 1) \succ (c, 1),$$

with any monotone completion. At every congestion vector containing post a , both agents demand a , whose capacity is one; hence the deficit is one. If a is empty, both still demand the empty post at quote $(a, 1)$ and the deficit is two. Therefore $\delta^* = 1$.

An assignment using posts a and b and an assignment using posts a and c are both least-coercive. Keep the same agent at a in the two assignments. The former Pareto dominates the latter because the other agent receives $(b, 1)$ instead of $(c, 1)$. Thus the least-coercive set can contain Pareto-dominated outcomes and can have different congestion and welfare profiles.

A secondary refinement can discard outcomes Pareto dominated by another least-coercive outcome; such a refinement is nonempty because the set is finite. This does not by itself establish global Pareto efficiency. More fundamentally, the binary deficit treats every violation equally. One could instead use rank loss, lexicographic regret, or cardinal welfare loss, but those alternatives add normative structure not present in the original ordinal model.

The results therefore settle a precise mathematical relaxation rather than uniquely resolve the authors' question. Their open problem asks for the appropriate competitive-fairness concept, which is partly axiomatic and normative. The present paper establishes the following defensible conclusions:

1. The original virtual-replacement quote $s_a \vee 1$ need not be altered by lower bounds; what fails is universal satisfaction of demand.
2. Fixed exemptions of exactly ℓ_a agents are mathematically insufficient in the presence of upper bounds elsewhere.
3. The smallest number of demand violations is well defined, always attained, and exactly equal to a capacitated Hall deficiency at each congestion vector.
4. The feasibility-forced population of a post set B is $\max\{\ell(B), n - u(\mathcal{A} \setminus B)\}$, so upper and lower bounds must be analyzed jointly.
5. A fractional least-coercive solution always exists and can be implemented by a floor/ceiling lottery.

Final classification. This is genuine and rigorous partial progress directly on the Bogomolnaia–Moulin lower-bound question. It is not a complete solution because it neither axiomatizes the deficit criterion uniquely nor recovers uniqueness and efficiency in general.

References

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